Signal, Image and Video Processing (2026) 20:286
https://doi.org/10.1007/s11760-026-05309-2

ORIGINAL PAPER

Comparative analysis of convolutional and vision transformer models for automated leukocyte classification enhanced by generative color augmentation

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Received: 22 October 2025 / Revised: 2 March 2026 / Accepted: 21 March 2026 / Published online: 24 April 2026

Abstract
Manual differential leukocyte counting is a critical yet time-consuming and observer-dependent process in clinical hematology. This study presents a comparative analysis of You Only Look Once v11 (YOLOv11) and Vision Transformer (ViT) architectures for the classification of 14 leukocyte types and artifacts using a private clinical dataset. We further investigated the impact of HistAuGAN, a domain-specific data augmentation strategy designed to simulate real-world staining variability. Across experimental settings, ViT models achieved higher overall performance than YOLOv11 variants, and the application of HistAuGAN led to systematic improvements in both architectural families. The best-performing configuration, `ViT-Base` trained on the HistAuGAN-augmented dataset, achieved a macro F1-score of 98.36% and an overall accuracy of 99.75% on the validation set. To assess generalization capacity, this configuration was additionally evaluated on the public PBC and LISC datasets, demonstrating meaningful cross-dataset performance without architectural modification. Model interpretability was examined through attention- and activation-based saliency analyses, indicating that predictions were primarily driven by morphologically relevant leukocyte regions rather than background structures. These findings suggest that combining global-context modeling with domain-informed augmentation provides a robust and clinically coherent framework for fine-grained leukocyte classification.

Keywords Artificial intelligence · Leukocytes · Image interpretation · Computer-assisted

1 Introduction

The differential count of leukocytes in peripheral blood is a cornerstone in the diagnosis and monitoring of a wide range of clinical conditions, including infections, inflammatory disorders, and hematological neoplasms such as leukemias [1, 2]. Despite advances in laboratory automation with hematology analyzers, the morphological analysis of blood smears under a microscope remains the indispensable gold standard for detecting immature, atypical, or reactive cells [3]. However, this manual process is inherently subjective, time-consuming, and susceptible to inter-observer variability, critically depending on the clinical analyst's expertise [4, 5].

In this context, artificial intelligence (AI), through deep learning algorithms, emerges as a promising tool to standardize, expedite, and increase the accuracy of leukocyte classification [6, 7]. Computer vision architectures, such as the Convolutional Neural Networks (CNNs) of the You Only Look Once (YOLO) family and the more recent Vision Transformers (ViT), have shown great potential in medical image analysis [8, 9]. Nevertheless, the literature still lacks studies that conduct a systematic and direct comparison between these different architectures on the same dataset that reflects the morphological diversity of a real clinical environment. Moreover, comparative studies frequently emphasize predictive accuracy while overlooking aspects such as computational efficiency, interpretability, and

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cross-dataset generalization, which are critical for real-world clinical deployment.

Additionally, a persistent challenge in applying AI in hematology is the staining variability between slides, a common artifact in laboratory practice [10]. Advanced data augmentation techniques, such as HistAuGAN, based on Generative Adversarial Networks (GANs) for style normalization, have been proposed as a solution, but their impact on the performance of leukocyte classifiers is still underexplored, especially when applied to multiple model architectures [11, 12]. Since color distribution and staining consistency directly influence morphological feature extraction, evaluating generative normalization strategies across distinct architectural paradigms becomes particularly relevant.

This study addresses these gaps by conducting a rigorous comparative analysis between models from the YOLOv11 and Vision Transformer families for the classification of 14 leukocyte types and artifacts using digital images obtained from a clinical laboratory. Beyond conventional accuracy-based evaluation, the study investigates the effects of domain-specific data augmentation using HistAuGAN, analyzes model interpretability through attention- and activation-based visualizations, evaluates cross-dataset generalization via linear probing on public benchmarks, and compares computational efficiency in terms of parameter count and inference latency under identical hardware conditions.

Accordingly, the objectives of this work are: (1) to compare the quantitative performance of YOLOv11 and ViT architectures across multiple model scales for leukocyte classification; (2) to evaluate the impact of HistAuGAN augmentation on predictive performance and robustness; (3) to qualitatively examine model interpretability for the best-performing configuration through attention-based visualization, assessing whether predictions are driven by morphologically relevant leukocyte regions; and (4) to examine the trade-offs between predictive accuracy and computational efficiency, discussing implications for real-time clinical deployment.

2 Related work

Deep learning has become the dominant paradigm for leukocyte classification, with numerous studies reporting high performance. This section summarizes the evolution of model architectures, common methodological strategies, and key limitations that motivate the present work.

2.1 Evolution of model architectures

The development of models for leukocyte analysis follows broader computer vision trends. Early works demonstrated the effectiveness of Convolutional Neural Networks (CNNs) using architectures such as AlexNet and VGG16 [13, 14]. Subsequently, deeper networks including ResNet and DenseNet became widely adopted as robust feature extractors and backbones for customized models [15, 16].

More recently, real-time object detection models from the YOLO (You Only Look Once) family have gained prominence. Variants such as YOLOv7, YOLOv8, and YOLOv11 reflect an increasing focus on efficiency and practical pipelines in which detection precedes classification [6, 17]. Lightweight architectures such as MobileNet and EfficientNet have also been explored to balance accuracy and computational cost [11, 18], highlighting the growing importance of architectural optimization and performance—efficiency trade-offs.

In parallel, Vision Transformers (ViT) have emerged as an alternative to CNN-based models. By modeling long-range dependencies through self-attention, ViTs have shown competitive or superior performance in classification tasks [19]. However, their higher parameter counts and computational demands raise questions regarding real-time feasibility compared to optimized convolutional detectors.

2.2 Methodological strategies and datasets

Two main methodological strategies are commonly reported. The one-stage approach performs direct end-to-end classification of cell images using a single model [20]. The two-stage approach first applies detection or segmentation (often using YOLO variants) to isolate the leukocyte Region of Interest (ROI), followed by dedicated classification [21, 22]. This separation may improve performance by restricting the classifier to relevant morphological features.

To enhance robustness, researchers employ attention mechanisms to emphasize discriminative regions [23] and Generative Adversarial Networks (GANs) to mitigate class imbalance and increase variability [21, 24]. Nevertheless, relatively few studies systematically assess whether predictions rely on clinically meaningful structures, reinforcing the need for interpretability analyses.

Most evidence in this field is derived from public datasets such as `BCCD`, `LISC`, `PBC`, and `Raabin-WBC`. While these benchmarks enable comparison across studies, differences in staining protocols, acquisition devices, and illumination conditions introduce domain shifts that may limit generalization. Cross-dataset validation remains underreported.

2.3 Remaining gaps and our contribution

Despite frequently reported accuracies exceeding 95%, heterogeneity in preprocessing, evaluation protocols, and limited class diversity complicate direct comparison and restrict clinical applicability [25].

Moreover, many works restrict classification to 4 or 5 mature leukocyte types, overlooking immature and atypical cells (e.g., blasts and promyelocytes) essential for diagnosing hematologic malignancies.

This study addresses these gaps through a rigorous comparison between YOLOv11 and Vision Transformer architectures on a clinically realistic dataset comprising 14 leukocyte classes. Beyond accuracy, we incorporate saliency-based interpretability analysis, cross-dataset generalization experiments, and a formal evaluation of computational efficiency, providing a comprehensive and deployment-oriented assessment framework.

3 Methods

This section describes the complete experimental protocol, from dataset acquisition and annotation to model training, interpretability analysis, and cross-dataset evaluation Fig 1.

3.1 Dataset and annotation

The image dataset used in this research was obtained through a collaboration with the clinical analysis laboratory at the Universidade do Vale do Itajaí (UNIVALI), Brazil. The data was formally provided for this project, which is affiliated with the Universidade Federal de Santa Catarina (UFSC). The dataset consists of anonymized digital leukocyte images captured by a CellaVision DM96 automated cell morphology system. The study was approved by the UFSC Research Ethics Committee (CAAE: 83684524.7.1001.0121).

The original dataset comprised 4,471 images. To ensure high-quality labels for supervised learning, a rigorous annotation protocol was implemented on the Roboflow platform. Initially, two experienced clinical analysts independently annotated the entire dataset. Subsequently, a third specialist acted as an arbitrator to resolve any discrepancies and validate the final labels. This process resulted in a final dataset of 5,011 annotations distributed across 14 distinct classes, as detailed in Table 1. Some images contain multiple annotations.

Table 1 Class distribution

Class	N	%
Segmented Neutrophil	1752	34.96
Lymphocyte	1033	20.61
Blast	438	8.74
Cellular Debris	378	7.54
Band Neutrophil	321	6.41
Atypical Lymphocyte	227	4.53
Monocyte	213	4.25
Eosinophil	203	4.05
Metamyelocyte	171	3.41
Myelocyte	97	1.94
Promyelocyte	92	1.84
Artifact	60	1.20
Erythroblast	18	0.36
Basophil	8	0.16
Total	5011	100

Table 2 Standard data augmentation techniques and parameters

Technique	Parameters Applied
Flip	Horizontal and vertical
Rotate	90° clockwise and counter-clockwise
Crop	Random zoom between 0% and 20%
Shear	±10° on horizontal and vertical axes
Saturation	Variation between -25% and +25%
Brightness	Variation between -16% and +16%
Exposure	Variation between -10% and +10%
Blur	Up to 2.5 pixels
Noise	Add noise to up to 0.1% of pixels

3.2 Data Pre-processing and augmentation

All images were pre-processed before training. They were first auto-oriented based on EXIF metadata and then resized to the input dimensions required by each architecture: 640 × 640 pixels for YOLO models and 384×384 pixels for Vision Transformer (ViT) models.

Two distinct data augmentation strategies were employed. First, a pipeline of standard augmentation techniques was applied to the original training images to artificially increase the dataset's size and diversity, which helps prevent overfitting. The set of stochastic transformations used, including flips, rotations, and color adjustments, is summarized in Table 2.

After applying standard augmentation, the dataset was split into training (80%), validation (10%), and test (10%) sets, as shown in Table 3.

To mitigate stain variability, HistAuGAN [12] was applied to the training set. This GAN-based approach disentangles morphology from staining attributes, generating realistic color variations while preserving structural information. The training set was expanded from 9,320 to 55,920 images, while validation and test sets remained unchanged.

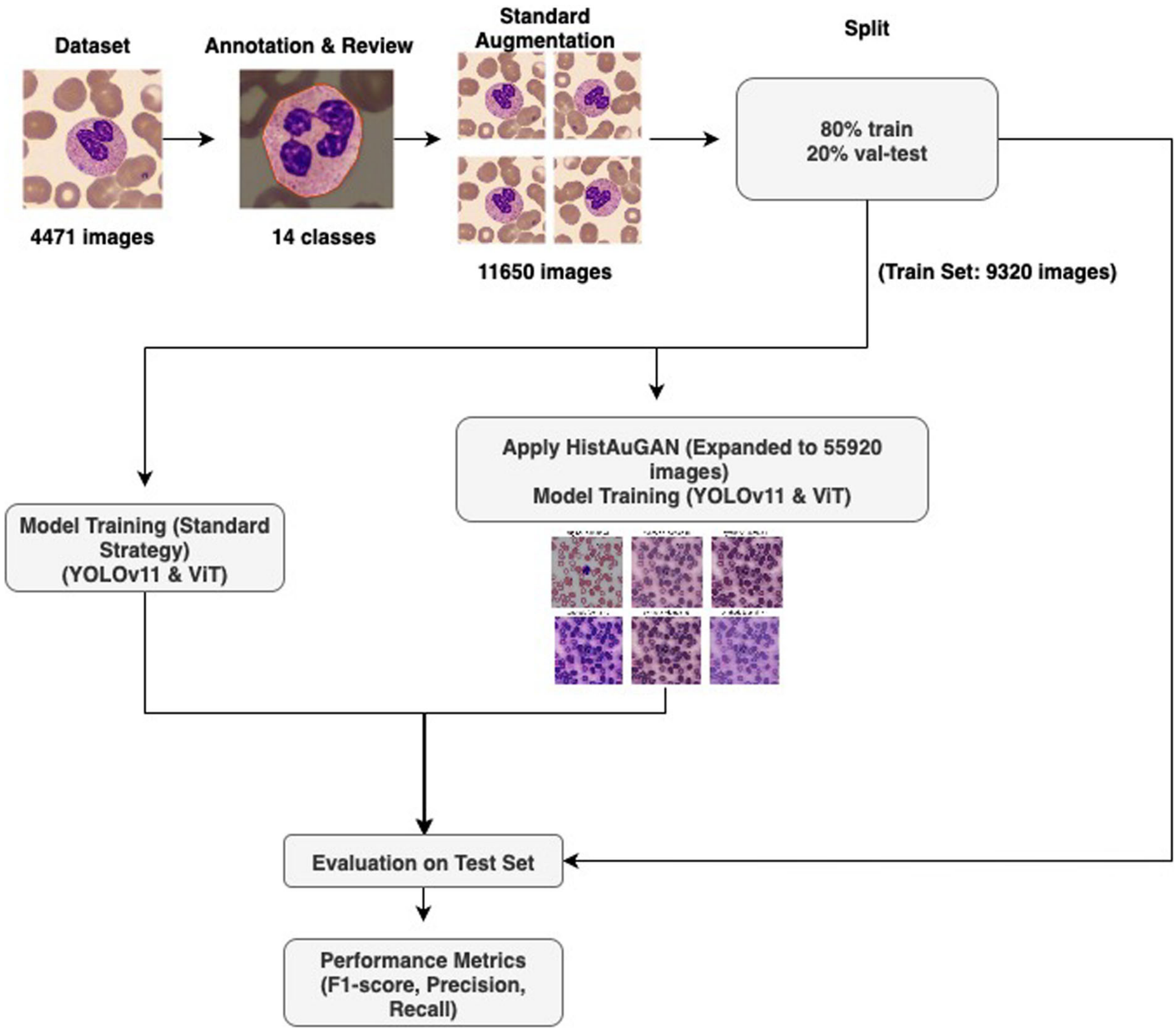

Fig. 1 Overview of the experimental pipeline. Two training strategies were evaluated (standard augmentation and HistAuGAN expansion). Models were assessed on a held-out test set and further analyzed in terms of interpretability, cross-dataset generalization, and computational efficiency

Table 3 Dataset split into training, validation, and test sets

Set	Proportion (%)	Number of Images
Training	80	9,320
Validation	10	1,165
Test	10	1,166

3.3 Model architectures

To provide a comprehensive comparison, we evaluated models from two distinct paradigms: Convolutional Neural Networks (CNNs), represented by the YOLO family, and Vision Transformers (ViTs).

3.3.1 CNN architecture (YOLO Family)

YOLO models are CNN-based detectors that perform feature extraction through stacked convolutional layers followed by prediction heads. Although originally designed for object detection, their backbone networks act as powerful feature extractors for classification tasks. We evaluated three variants: `YOLOv11-Nano`, `YOLOv11-Small`, and `YOLOv11-Medium` [8].

From a computational standpoint, `YOLOv11-Medium` contains 20.06 million parameters and requires 34.12 GFLOPs per forward pass (input resolution 640×640), making it considerably lighter than transformer-based alternatives. Its architecture is optimized for real-time inference, which is particularly relevant for deployment in laboratory environments.

3.3.2 Vision transformer (ViT) architecture

Vision Transformers divide an image into fixed-size patches, project them into embeddings, and process them using a Transformer Encoder with multi-head self-attention. This mechanism enables global contextual modeling across the entire image. Final classification is performed via an MLP head. We evaluated two variants: `ViT-Small` (`vit_small_patch16_384`) and `ViT-Base` (`vit_base_patch16_384`) [9].

`ViT-Base` contains 85.66 million parameters and requires 197.57 GFLOPs per forward pass (input resolution 384×384), resulting in substantially higher theoretical computational complexity compared to `YOLOv11-Medium`. However, transformer architectures benefit from efficient parallelization on modern GPUs, partially mitigating latency differences in practice.

3.3.3 HistAuGAN for data augmentation

To address stain variability, we employed `HistAuGAN`, a GAN-based augmentation method that disentangles morphological content from staining style [12]. The model uses a domain-invariant content encoder (E_c) and a domain-specific attribute encoder (E_a), enabling the generator (G) to synthesize realistic stain variations while preserving cellular structure. This strategy increases color diversity without altering diagnostically relevant morphology.

3.4 Evaluation metrics

Model performance was evaluated on the held-out test set using Accuracy, Precision, Recall, and macro-averaged F1-score to account for class imbalance.

3.5 Cross-dataset generalization

To evaluate robustness under domain shift, we conducted cross-dataset experiments using the public PBC and LISC benchmarks. A linear probing strategy was adopted: the pre-trained backbone of the best-performing model was frozen, and only the classification head was trained on each target dataset.

This protocol enables assessment of representation quality without full fine-tuning, providing a more stringent evaluation of generalization across variations in staining, illumination, and acquisition conditions.

3.6 Computational efficiency analysis

To analyze practical deployment feasibility, we compared the computational efficiency of `YOLOv11-Medium` and `ViT-Base` models. The number of trainable parameters was computed directly from the model weights.

Inference latency was measured under identical hardware conditions (NVIDIA H100 GPU, FP32 precision, batch size = 1). For each model, a warm-up phase of 20 forward passes was performed, followed by 100 timed forward passes using input resolution 640×640 (YOLO) and 384×384 (ViT). The average inference time per image (milliseconds) and corresponding frames per second (FPS) were then computed.

These measurements provide an empirical comparison of real-time suitability between convolutional and transformer-based architectures.

4 Results

This section presents the quantitative results of the YOLOv11 and ViT models on the leukocyte classification task. The performance was evaluated on a held-out test set under two data augmentation strategies. Group A denotes models trained with standard augmentation, while Group B includes HistAuGAN. All metrics were calculated using macro averages to account for the 14-class imbalance.

4.1 Comparative performance analysis

Table 4 summarizes the performance of Group A. The `ViT-Base` model (A5) outperformed all YOLOv11 variants, achieving an F1-score of 97.74%. The transition to

Table 4 Performance with standard augmentation (Group A)

ID	Architecture	Acc (%)	F1 (%)
A1	YOLOv11-N	96.49	94.53
A2	YOLOv11-S	97.41	95.12
A3	YOLOv11-M	96.71	91.70
A4	ViT-Small	99.39	95.92
A5	**ViT-Base**	**99.66**	**97.74**

The bold values indicate the best performing models within each group, corresponding to the highest accuracy and F1-score achieved.

Table 5 Performance with HistAuGAN (Group B)

ID	Architecture	Acc (%)	F1 (%)
B1	YOLOv11-N	97.73	95.54
B2	YOLOv11-S	98.11	96.89
B3	YOLOv11-M	98.64	97.32
B4	ViT-Small	99.65	97.66
B5	**ViT-Base**	**99.75**	**98.36**

The bold values indicate the best performing models within each group, corresponding to the highest accuracy and F1-score achieved.

Group B (Table 5) shows a universal performance gain. The top-performing configuration, `ViT-Base` trained with HistAuGAN (B5), achieved a macro F1-score of 98.36% and an overall accuracy of 99.75%. Detailed confusion matrices and per-class performance metrics for all evaluated models are provided in Online Resource 1.

4.2 Model interpretability and efficiency

To address clinical interpretability, Attention Maps were generated for the `ViT-Base` model. As illustrated in Fig. 2, the self-attention mechanism qualitatively suggests prioritization of morphologically relevant regions, such as nuclear lobulation and cytoplasmic granulation, while successfully ignoring background artifacts. Additional attention rollout visualizations and saliency analyses are available in Online Resource 1.

Regarding computational efficiency (Table 6), while `YOLOv11-Medium` offers higher throughput, `ViT-Base` maintains a competitive inference latency of 9.54 ms on the NVIDIA H100 platform (104 FPS). This latency is compatible with real-time laboratory workflows under typical throughput conditions.

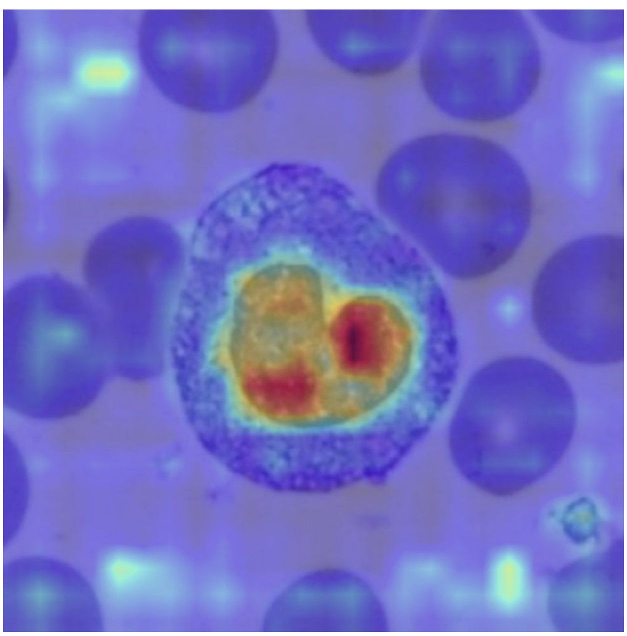

Fig. 2 ViT-Base Attention Maps highlighting the focus on the leukocyte nucleus and cytoplasm

Table 6 Latency (NVIDIA H100), FLOPs, and Parameters

Model	Params	FLOPs	Latency	FPS
YOLOv11-M	20.0M	34.1G	6.62ms	151
ViT-Base	85.6M	197.5G	9.54ms	104

Table 7 Generalization via Linear Probing (5-fold CV)

Dataset	Acc. (%)	Prec. (%)	F1 (%)
PBC	98.88 ± 0.21	98.68 ± 0.23	98.67 ± 0.23
LISC	91.73 ± 1.85	91.75 ± 2.31	91.06 ± 2.37

4.3 Generalization on public benchmarks

To ensure the findings are not overfitted to a specific laboratory protocol, the `ViT-Base` backbone (frozen) was evaluated via Linear Probing on two public datasets: PBC and LISC (Table 7). The model achieved a macro F1-score of 98.67% on PBC and 91.06% on LISC. These results indicate robustness to domain shifts in staining, lighting, and acquisition conditions.

5 Discussion

This study compared Transformer-based (ViT) and CNN-based (YOLOv11) architectures for leukocyte classification and evaluated the impact of stain-aware augmentation using HistAuGAN. Two main findings emerged: (1) ViT models achieved superior classification performance under the evaluated conditions, and (2) HistAuGAN consistently improved generalization across all architectures.

5.1 Architectural considerations and efficiency trade-off

The superior performance of `ViT-Base` is attributed to its global self-attention mechanism, which captures spatially distributed morphological cues such as nuclear structure and cytoplasmic granularity.

However, despite the predictive advantage of `ViT-Base`, `YOLOv11-Medium` demonstrated substantially lower computational complexity and higher throughput. Deployment decisions should therefore consider the trade-off between marginal accuracy gains and inference efficiency, particularly in high-volume laboratory settings.

5.2 Impact of stain-aware data augmentation

HistAuGAN yielded consistent performance gains across all tested models, supporting the hypothesis that stain variability is a major factor limiting generalization in hematological imaging. In routine laboratory practice, variations in staining protocols and acquisition conditions may introduce color shifts that are diagnostically irrelevant. By exposing models to realistic stain variations while preserving morphology, HistAuGAN encourages learning of invariant structural features rather than spurious color correlations. Notably, the F1-score of `YOLOv11-Medium` improved from 91.70% to 97.32%, highlighting the effectiveness of stain-aware augmentation.

5.3 Relation to prior work

Our findings align with the broader scope of architectural optimization in leukocyte classification. In particular, the superiority of Transformer-based models observed here corroborates the results of Mohamad Abou Ali et al. (2023) [19], who identified Vision Transformers (ViT) as the most effective approach on the PBC and BCCD datasets, achieving 100% accuracy on PBC. Similarly, our ViT-Base model outperformed CNN-based architectures under both internal evaluation and cross-dataset linear probing, reinforcing the benefit of global self-attention mechanisms for hematological image analysis.

Recent studies have explored complementary optimization strategies to improve robustness and efficiency. Hita et al. (2025) [26] introduced Bayesian CNNs for uncertainty quantification, Rodrigues et al. (2022) [27] applied Genetic Algorithms for automated hyperparameter tuning, Das and Meher (2021) [28] proposed hybrid MobileNetV2–ResNet18 architectures for efficiency, and Saikia et al. (2024) [29] developed BSNEU-net with Union Blocks and switchable normalization to handle heterogeneous datasets. While many of these approaches focus on binary or limited-class problems, our work extends architectural optimization to a 14-class diagnostic scenario under realistic stain variability.

When compared to prior reports on external datasets, our linear probing results remain competitive despite the stricter evaluation protocol. On LISC, Khan et al. (2025) [30] reported 98.78% accuracy using YOLOv8 and Manthouri et al. (2022) [31] achieved 97.14% with SIFT+CNN. On PBC, Mohamad Abou Ali et al. (2023) [19] achieved 100% accuracy with ViT, and Khan et al. (2025) [30] reported 99.18% using YOLOv8. Notably, these studies optimized models directly on the target datasets, whereas our backbone-freezing strategy evaluates feature transferability without full fine-tuning, providing a more stringent assessment of cross-domain generalization.

5.4 Limitations and generalization analysis

Per-class analysis revealed remaining challenges. `ViT-Base` showed reduced performance in the *Artifact* class, whereas `YOLOv11-Medium` struggled within the granulocytic maturation continuum (e.g., *Band Neutrophil* vs. *Metamyelocyte*), reflecting the intrinsic morphological overlap between adjacent differentiation stages.

Although the primary dataset originated from a single institution, external evaluation via linear probing on PBC and LISC provided insight into cross-domain robustness. The observed performance differences between datasets likely stem from acquisition heterogeneity rather than architectural limitations. PBC images are highly standardized and high-resolution, whereas LISC presents lower resolution, greater background variability, and smaller or more spatially distant cells. These factors increase domain shift and highlight the sensitivity of learned representations to imaging conditions.

Furthermore, the linear probing protocol, while suitable for evaluating feature transferability, does not fully capture the potential performance achievable through end-to-end fine-tuning on multi-center data. Future work should therefore investigate multi-institutional training strategies and knowledge distillation to transfer transformer-level representations into lightweight models suitable for resource-constrained environments.

Overall, the combination of global-context modeling and stain-aware augmentation provides a scalable framework for multi-class leukocyte classification, though broader validation across acquisition settings remains necessary.

6 Conclusion

This study presented a comparative evaluation of YOLOv11 and Vision Transformer architectures for 14-class leukocyte classification under realistic stain variability conditions. `ViT-Base` achieved the highest macro F1-score (98.36%) when combined with HistAuGAN, while `YOLOv11-Medium` offered competitive performance with lower computational complexity and higher throughput. External evaluation on PBC and LISC confirmed meaningful cross-domain generalization.

These findings indicate that combining global-context modeling with stain-aware augmentation provides a robust framework for fine-grained leukocyte analysis, while deployment decisions should balance predictive gains against computational constraints.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11760-026-05309-2.

Acknowledgements The authors thank LEAC/UNIVALI for providing the clinical dataset and UFSC for access to high-performance computing infrastructure (NVIDIA H100).

Author Contributions J.K. and A.G.S. conceived and designed the study. J.K. performed the experiments, conducted the data analysis, prepared all figures, and wrote the original draft of the manuscript. A.G.S. provided supervision and resources. Both authors critically reviewed and edited the manuscript and approved the final version.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). No funding was received for this study.

Data Availability The dataset generated and analyzed during the current study is not publicly available due to patient privacy and the terms of the ethical approval (UFSC Research Ethics Committee, CAAE: 83684524.7.1001.0121) under which the data was provided by a partner institution. The data may be made available from the corresponding author upon reasonable request and with the permission of the data-providing institution.

Code Availability Available from the corresponding author upon reasonable request.

Declarations

Conflicts of interest The authors declare no competing interests.

Ethics approval Approved by the Research Ethics Committee of the Federal University of Santa Catarina (CAAE: 83684524.7.1001.0121; Opinion No. 7.253.688), in accordance with the Helsinki Declaration.

Consent to participate and publication Not applicable (retrospective study using de-identified images).

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